

Techno-Economic Analysis of Hybrid Power Supply Scenarios for a Resort in Egypt

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Abstract— Numerous remote areas throughout the globe have obstacles in acquiring electric energy due to the challenging process of reaching the national grid. Consequently, energy customers in these regions have turned to a range of energy sources, including traditional ones like diesel generators (DGs), as well as renewable options like solar photovoltaic (PV), wind, or wave power, while waiting for a connection to the grid. This paper presents an analysis of seven power supply scenarios, examining their techno-economic feasibility, with the aim of selecting the most optimal energy solution for a resort located in Marsa Alam, Egypt. The examined scenarios include DG, PV, and grid connections, along with the potential inclusion of batteries, hydrogen electrolyzers, and fuel cells (FC). The Hybrid Optimization Model for Electrical Renewable (HOMER PRO) energy modeling and simulation software was used to assess the energy balance, economics, and environmental aspects of the different scenarios. The results derived from the simulations indicate that the grid connected system with solar energy and DGs is the most economically efficient option. However, it is worth noting that all the presented scenarios demonstrate technological feasibility and economic parity in the long run, although with distinct benefits and challenges.

Keywords—Hybrid power system, sustainability, Renewable energy, HOMER Pro, green hydrogen, Techno-Economic, Solar Energy, Remote Area, off grid, on grid

I. INTRODUCTION

Egypt is the most populated nation in the Middle East with 113 million people in 2023[1], faces significant energy challenges due to its expanding economy and rapid population growth. To address these challenges, International Renewable Energy Agency (IRENA) asserts that renewable energy (RE) represents a crucial solution, aligning with Egypt's goals for sustainable development and climate action [2].

The ministry of Electricity and Renewable Energy has established the Integrated Sustainable Energy Strategy (ISES 2035) that outlines Egypt's ambitious targets to

increase renewable energy's contribution to 20% by 2022 and 42% by 2035 [3]. In addition, Egypt announced many initiatives during the COP27 climate conference at Sharm El Sheikh in 2022 [4] such as the launch of the national strategy for green hydrogen, which underscore Egypt's commitment to a low-carbon future. Despite efforts to extend the national grid, many remote areas still rely on DGs, driving interest in RE adoption, specifically solar energy due to Egypt's strong solar irradiance for most of the year, also because the unstable diesel prices and expensive transport costs of diesel to remote areas.

Various studies have provided comprehensive reviews of hybrid energy systems designed for remote and tourist-centric areas around the world such as, Abdelaal et al. in [5] who investigated a hybrid energy system for a resort close to the Sokhna coast in Egypt, to achieve an optimal balance of Grid, PV, wind, diesel, and battery banks. It was found that on-grid hybrid system minimizes energy costs while ensuring reliability as its energy cost is \$0.0443/kWh compared to \$0.134/kWh for the off-grid hybrid system. Using HOMER PRO software, Shezan et al. in [6] focused on a similar system for a resort on Penang Island in Malaysia. The results showed a 40.8% reduction in emissions and a 50% reduction in net present cost (NPC) for hybrid RE systems compared to traditional power plants. In another research, Fofang and Tanyi conducted a techno-economic analysis for a remote area in Cameroon [7], confirming the viability of off-grid systems using battery bank with \$0.268/kWh as a Levelized cost of energy (LCOE). Hossain et al. [8] designed a hybrid RE system for a resort in Malaysia using HOMER PRO, consisting of PV, wind, DG, and batteries, with LCOE of \$0.28/kWh. By comparing several cost-effective scenarios for modelling the available power resources (PV, Wind, Biomass, and Fuel Cell) using HOMER PRO, Jahangiri et al. [9] demonstrated significant reductions in emissions and overall cost by using solar and biomass to supply off-grid popular tourist site in Iran. Furthermore in [10], a 1.3 MW hybrid energy generation system including PV, wind, biodiesel, and DG was designed by using the most economical scenario by HOMER PRO for Saint-Martin Island, a tourist destination in Bangladesh with LCOE of \$0.193/kWh. Ghenai and Bettayeb [11] focused on a standalone PV/fuel cell/DG system for Sharjah university in the UAE, optimizing LCOE by 73%. Li [12] investigated

hybrid systems including PV/FC/battery, for 100 residential buildings in China where the most effective system was hybrid PV and battery, with LCOE of \$1.536/kWh. In addition, the most suitable FC, PV, and battery system has LCOE of \$1.553/kWh. Cozzolino et al. [13] constructed a hybrid wind/solar/FC/battery power system for artificial islands in Italy where HOMER PRO software is used to size the energy system components. LCOE that was discovered to be the lowest was 0.522 €/kWh.

According to previous literature, HOMER PRO has been used to simulate and analyze various studies on hybrid energy systems for remote and tourist-centric locations worldwide. That's why HOMER PRO is used in this article to perform simulation for the selected resort in Marsa Alam, Egypt scenarios, considering factors like load profile, excess electricity, NPC, LCOE and carbon emissions.

II. CASE STUDY DESCRIPTION

A. Location

The investigated resort [14] is located in Marsa Alam, Egypt, small town near the Red Sea in Egypt's Eastern Desert, located 790 km south of Cairo, 300 km from Hurghada, and about 200 km away from Sudan borders. The exact location of the resort under study is at (24°55'05.53"N 34°58'43.95"E) in the southern part of Marsa Alam, Red Sea, Egypt.

B. Description of Resort's Power System

The power system for the resort, which consists of five hotels and staff housing facilities, relied primarily on (DGs) to produce electricity until 2018. Due to high operating costs, pollution and noise, the owners of the resort decided to embark on a transformative journey, and carried out energy audits, which led to the use of LED lighting and solar water heaters, resulting in a notable 40% reduction in energy consumption.

Subsequently, at the end of the first quarter of 2019, a 2.5 MW solar power plant was installed as shown in Fig 1 through two phases, which significantly reduced fuel consumption and CO₂ emissions, prompting the resort to operate as an isolated hybrid grid during daylight hours.



Fig 1: Resort's 2.5MW Solar Plant

By the end of the second quarter of 2021, due to the unpredictable and intermittent nature of solar energy, requirements of stability enhancement necessitated connecting the resort to the closest national grid [15].

C. Load Profile

The load profile refers to the electrical demand and consumption at the resort. The resort is a tourist destination with Air Conditioning being the dominant loads due to the geographical location of the resort resulting in an increased energy demands during summer months. In general, the increases of load are observed during the period from April to September, which is the high season of tourism.

D. Temperature Profile and Solar Radiation

The annual average daily temperature is 25.07°C with August having the highest average daily temperature which is 32.13°C while January has the lowest average daily temperature which is 16.81°C. The annual average value for global horizontal radiation (GHI) considering National Renewable Energy Laboratory (NREL) database is 6.22 kWh/m²/day such that June has the highest radiation which is 8.036 kWh/m² per day while December has the lowest radiation which is 4.207 kWh/m² per day [16].

III. METHODOLOGY

This study utilizes HOMER PRO to find the best available scenario by comparing NPC and LCOE at the current project phase with the proposed designs. HOMER PRO is a micro power optimization model, that simplifies the task of evaluating designs of both off-grid and grid-connected power systems for a variety of applications. The HOMER Software is usually used for designing and analyzing hybrid power systems, which contain a mixture of conventional generators, cogeneration, wind turbines, PV, hydropower, batteries, Fuel Cell, biomass, and other inputs [17].

In order to simulate multiple system configurations, HOMER PRO uses inputs such as various technology options, component costs, resource availability, manufacturer's data etc. The software HOMER generates simulation results in a wide range of tables and graphs, making it easier to compare configurations and assess their relative technical and economic benefits. It can choose the least expensive source to meet the load according to load serve policies.

IV. POWER SYSTEM COMPONENTS

A. Diesel Generator (DG)

The installed DG model is (CAT-C15-500kVA-50Hz-PP) with 400 kW capacity at 0.8 Power factor at 400 Volt and the generator speed of rotation is 1500 RPM. Fuel for this generator is Diesel with price 0.5 \$/liter. Capital cost of the generator is 100,000 \$, replacement cost is 75,000 \$, and the operating and maintenance cost is 2.5 \$/hour with 80,000-hour lifetime and 30% minimum load ratio [18].

B. Photovoltaic Cell (PV)

SUNTECH STP315-24/Vem solar panel poly-crystalline type module is installed as part of the power system of the resort. The given efficiency by the manufacturer is 16.2% and 25 years of lifetime. Capital cost of PV is 600 \$/kW, Replacement cost is 400 \$/kW, and the operating and maintenance cost is 5000 \$/year [19] as actual data collected from the resort.

C. Inverter

The utilized inverter is String inverter TRIO-50.0-TL-OUTD with MPPT, which has 98% efficiency with 15 years lifetime. The capital cost is 8.5k\$, replacement cost is 8k\$ without operation and maintenance cost [20].

D. Grid

The electricity distribution company uses net metering system which measures the difference between the amount of electricity the system generates (PV or DGs) throughout the month and the amount of electricity that the national grid supplies, then the resort pays fixed price for total result of consumption after net metering system calculation regardless of the peak time. The grid extension cost was about 0.9M\$ with 0.1 \$/kWh grid power price and 0.05 \$/kWh Grid Net Excess Price.

E. Battery

Battery bank consisting of Trojan SIND 06 610 batteries is used for off-grid operation due to their lower cost compared to Li-ion batteries. Trojan SIND 06 610 is a deep cycle lead acid type battery with a nominal capacity of 635 Ah (3.18 kWh) at a nominal voltage of 6V [21]. After surveying the battery market for the selected model, the capital cost is set at \$650, replacement cost for battery is assumed to be \$600 and operation and maintenance to be \$5/year/battery.

F. Electrolyzer

Without any carbon emissions, High-purity hydrogen and oxygen will be produced from water by electrochemical process via the electrolyser using direct electric current. In this study, an Proton Exchange Membrane (PEM) electrolyser is chosen for the simulations, lifetime depends on operating profile which set as 25-year, capital and replacement costs are set as 1500 \$/kW and O&M cost is set as 30\$/kW/year and efficiency of 80% [22, 23].

G. Hydrogen Tank

Metal cylinder pressure vessels fed by a compressor are the most common method of storing hydrogen since they are long-lasting and appropriate for small-scale uses [24]. The hydrogen tank's capital cost is estimated to be \$1200/kg, while its replacement, O&M, and lifetime costs are set at \$1100/kg, \$10/kg/year, and 25 years, respectively.

H. Fuel Cell (FC)

The process of converting the chemical energy of hydrogen into electricity and heat is mostly dependent on FC. In comparison between FC types, PEM type FCs provide lower weight, volume, temperature advantages along with high cell lifetime, electrical efficiency, quick start-up speeds, and quiet, (locally) emission-free electricity generation. PEM FC with 40000h lifetime is chosen for this case study and corresponding capital cost and replacement cost are assumed as 2500\$/kW and 2300\$/kW with O&M cost is 0.02\$/h [25].

V. ECONOMIC MODEL

HOMER PRO calculates NPC of each suggested power system and ranks them in descending order based on NPC. The NPC of a system is the present value of all

costs during its life minus revenues and present value of the system. It is an important indicator as it shows if the whole investment is profitable or not. The NPC is given by the formula below:

$$NPC = \frac{C_{ann_tot}}{CRF(i, N)} \quad (1)$$

Where C_{ann_tot} is Total annualized cost [\$/year] and $CRF(i, N)$ is capital recovery factor which can be calculated by the following Equation:

$$CRF(i, N) = \frac{i(1+i)^N}{(1+i)^N - 1} \quad (2)$$

Where N is Number of years and i is the real discount rate that can be calculated by HOMER uses the following equation:

$$i = \frac{i' - f}{1 + f} \quad (3)$$

Where i' is Nominal discount rate in (%) and f is the expected inflation rate.

Homer uses total annualized cost C_{ann_tot} to calculate LCOE using the following equation:

$$LCOE = \frac{C_{ann_total}}{E_{total}} \quad (4)$$

Where E_{total} is Annual total energy produced (kWh).

In addition to some of the financial factors, such as operating costs, NPC, LCOE, initial capital costs, and replacement costs of each system, this process also takes into account CO2 emissions, the production of each hybrid energy system, and excess energy.

The overall system architecture for the input data, which includes capital, replacement, operation, and maintenance (O&M) costs as well as lifetime, uses actual market prices as economic inputs in Table 1.

Table 1: Actual Economic inputs

Component	Lifetime	O&M Cost	Replacement Cost	Capital Cost	Type/Model
(DG)	80000 Hour	2.5 \$/h	75000 \$	100000 \$	CAT-C15-500kVA
PV	25 Year	5000 \$/Year	400 \$/Kw	600 \$/kw	SUNTECH STP315
Converter	15 Year	0	8000 \$	8500 \$	ABB TRIO-50.0-TL-OUTD
Electrolyser	25 Year	30 \$/kW	1500 \$/kW	1500 \$/kW	PEM
Hydrogen tank	25 Year	10 \$/kg/year	1200 \$/kg	1200 \$/kg	Metal cylinder vessels
Fuel Cell	15 Year	0.02 \$/h	2300 \$/kg	2500 \$/kg	PEM
Battery	4500 Cycle	5 \$/kW/Year	600 \$	650 \$	Lead Acid

VI. HYBRID POWER SYSTEM SCENARIOS

Case studies in this research contains seven simulated and analyzed scenarios by HOMER PRO software. A base (existing) scenario described the current hybrid solution in the resort in addition to additional six scenarios that are proposed and compared to the existing power scenario. The comparison's objective is to reach the least LCOE with stable performance and low carbon emissions.

The investigated and proposed power supply scenarios are outlined as follows:

- Case study 0: is the existing Scenario (base Case): (DG) – (PV) – Grid
- Case study 1: (DG) – (PV) – Batteries
- Case study 2: (DG) – Modified (PV) – Batteries

- Case study 3: (DG) – Modified (PV) – Grid
- Case study 4: (DG) – Modified (PV) – Grid – Batteries
- Case study 5: (DG) – (PV) – H₂
- Case study 6: (DG) – Modified (PV) – H₂

*Note: Modified (PV) means additional PV installed.

A. Case study 0: (DG) – (PV) – Grid

This case study is the existing power supply scenario and in fact represents the most recent phase. It is as shown in fig 2 and is considered as a hybrid power system with connection to the national grid.

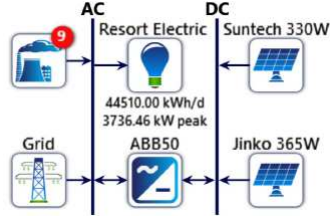


Fig 2. Schematic diagram of Existing Scenario (Case study 0)

Background history of power generation for the resort includes two phases that the resort's power system underwent before reaching the existing state, where the first phase lasted for 12 years from 2006 till 2018, the power system of the resort was standalone micro grid based on 11 Caterpillar Inc. generators as the main source of electricity without RE integration. Resort load demand was met by nine DGs with a total capacity of 3,600 kW out of the 11 available and the other two generators were served as backup to avoid any power supply shortage.

The second phase started from 2018 when resort's owner decided to reduce diesel usage because of logistical challenges, environmental concerns, and high costs. Accordingly, SUNTECH power 315STP315-24/Vdx 315W solar panels and Jinko Solar365JKM365M-72 365W solar panels were selected with a total capacity of 2.5 MW which is equipped with DC to AC converter module combined with DGs to serve for periods of prolonged lack of sunshine.

The interconnection of the national grid mitigated the use of DGs which was a top priority for the owner of the resort but on the other hand, the power plant in the resort could not completely abandon the use of generators due to the reliability of the network, which was evident in the first year of interconnection with the national grid. There were 8 grid outages with an average repair time of 8 hours so they used generators to meet the load at these times.

B. Case study 1: (DG) – (PV) – Batteries

The first case study suggests replacing the national grid with batteries. Therefore, this scenario has base case components which are already installed like DGs, the same capacity of PV panels (2.5 MW) and inverters in addition to batteries as shown in Fig. 3.

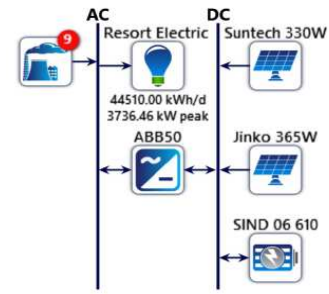


Fig 3. Schematic diagram of Case study 1

C. Case study 2: (DG) – Modified (PV) – Batteries

Case study 2, is considered as a modified version for the latest one, the estimated additional power capacity of PV after checking the physical space availability is 1.5 MW to reach 4 MW as a total energy capacity of PV. Fig 4 shows the schematic diagram of case 2 through HOMER PRO software to reach the optimal solution.

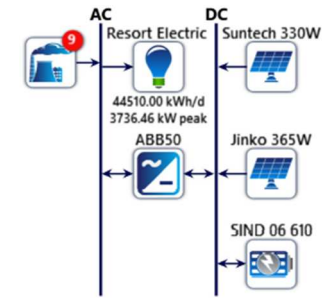


Fig 4. Schematic diagram of Case study 2

D. Case study 3: (DG) – Modified (PV) – Grid

This case, as shown in Fig 5, is considered as an adjustment of existing scenario (case 0) by increasing capacity of the PV to reach 4 MW instead of 2.5 MW.

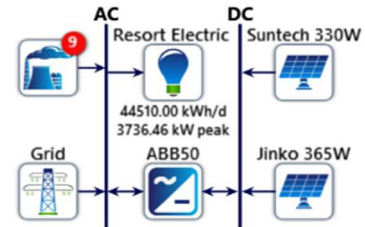


Fig 5. Schematic diagram of Case study 3

E. Case study 4: (DG) – Modified (PV) – Grid – Batteries

For further optimization of case study 3, batteries are proposed to be added to the system as shown in Fig 6, which may reduce NPC and LCOE.

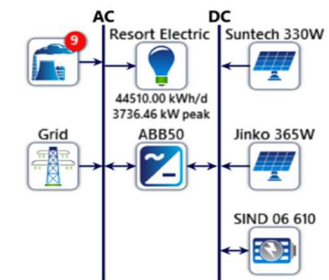


Fig 6. Schematic diagram of Case study 4

F. Case study 5: (DG) – (PV) – H₂

The proposed system in this scenario is designed to improve the off-grid hybrid system using hydrogen system which stores excess energy by converting electricity into hydrogen through

an electrolyser with rated capacity 850kW, then the produced hydrogen is stored in a hydrogen tank with capacity 150 kg, finally the fuel cell with rated capacity 250 kW generates power again using the stored hydrogen when needed. Fig 7. shows the schematic of this scenario in HOMER PRO Software, such that this scenario contains 8 DGs and 2.5 MW of PV in addition to hydrogen system as a kind of energy storage system.

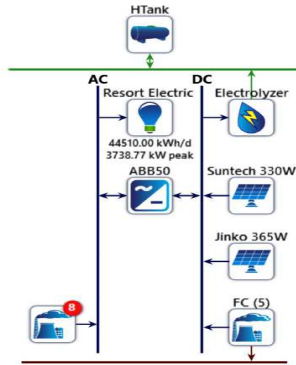


Fig 7. Schematic diagram of Case study 5

G. Case study 6: (DG) – Modified (PV) – H₂

In this scenario, PV capacity is designed to vary from 2.5 MW to 4 MW, size of the electrolyser also varies from 850 kW to 2 MW, hydrogen tank from 100 kg to 300 kg and two fuel cells with a capacity of 250 kW each as shown in Fig 8.

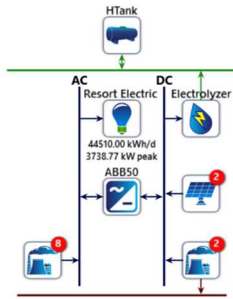


Fig 8. Schematic diagram of Case study 6

VII. SIMULATION RESULTS AND DISCUSSION

To meet the electricity demands of a resort in Egypt, this study attempts to evaluate the feasibility and financial viability of utilizing the most efficient hybrid energy scenario. HOMER PRO conducted simulation for each potential hybrid energy system in order to evaluate their features and economic viability. All the proposed hybrid energy scenarios (cases 1 to 6) were evaluated and compared with the base case systems to reach the optimal design. Simulation results for the various hybrid energy systems are discussed in the subsequent sections.

A. Yearly Production and Energy Consumption

Total power production to meet the load at case 0 is 16,500 Mw, which contained three main components as shown in Fig 9, Grid had the greatest share in the electrical production with about (69%), PV (28.5%) and (2.43%) for DGs which only use at outage times. In case 1, the following generation sources (nine DGs with 71.5% and PV with 28.5%) serves the electrical load. Case 2 was implemented with a renewable adoption rate of 42.9% instead of 28.5% in the previous case, so the power generated from DGs has been reduced to 57.1%, while storage system's nominal capacity is 2,163 kWh and the annual throughput is 118,800 kWh/yr. In case 3, HOMER PRO suggested decreasing the number of DGs to only six, instead of nine in case study 0. The optimal

results indicated 55.4% of the total power usage is shared by national grid, while PV share with 42.8% and DGs for grid outage times only. While case 4 suggested 5 DGs instead of 6, with a nominal capacity of 91.4 kWh and annual throughput of the battery storage system is 861 kWh/year, the optimal results also suggest using the same percentage of the national grid 55.4% of the total power usage like case 3, PV at 42.8%, while the main difference is batteries and DGs together for grid outage times not DGs only. Case 5 consists of an electrolyzer with a 0.85 MW capacity that produces 33,574 kg of hydrogen annually by receiving 1,558 MWh/yr; a 150 kg capacity hydrogen tank; an FC with a 250kW rated capacity that produces 742.5 MWh/yr and a 227.5 MWh/yr thermal production. On the other hand, DGs have the highest share ratio with 69.88% followed by PV with 26% and FC with 4.12%. Case 6 shows that DGs have the highest share ratio of power generation at 54.2%, then PV at 39.3%. It also includes a 1.5 MW electrolyzer that receives 2,642 MWh annually to produce 56,932 kg of hydrogen, a 250 kg hydrogen tank, and two 500 kW FCs that produce 6.55% of the total power generation and 385 MWh of thermal production annually.

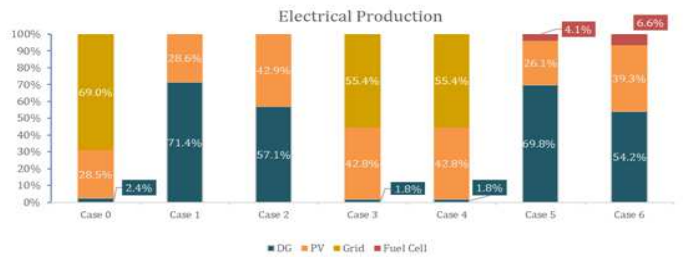


Fig 9. Electrical production resources percentage

The energy consumption for the previous scenarios is shown in Fig. 10. Case 0 consumes 16,400 MWh/yr, mainly for AC load (99.1%) and a small portion for grid sales (0.935%). Cases 1 and 2 show similar AC load consumption. Conversely, Case 3, as on-grid scenario, consumes 17,383 MWh/yr, with 93.5% for AC load and 6.54% for grid sales. Case 4 similar to previous case with 93.4% for AC load and 6.61% for grid sales. Case 5, first hydrogen-based scenario, consumes 17,802 MWh/yr, with 91.2% for AC load and 8.8% as electrolyser input energy. Finally, Case 6, also hydrogen-based, consumes 18,888 MWh/yr, with 86% for AC load and 14% for electrolyser input energy.

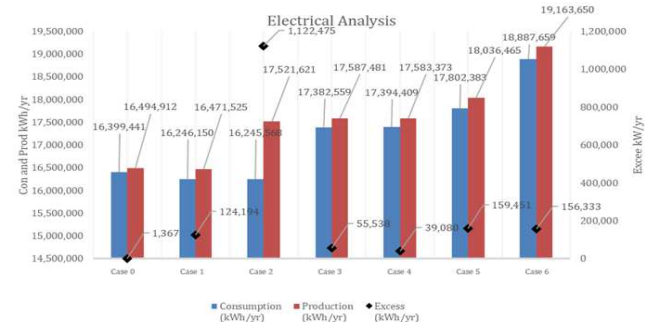


Fig 10. Comparison of energy production, consumption, and excess

B. Cost Analysis

To identify the most economically advantageous scenario for producing electricity, Table 2 compares proposed cases with the existing one. It shows that the fifth scenario (DG-PV-H₂) has the lowest capital cost of the other scenarios at 2,716,667\$, while the third case (DG-Modified PV-Grid) is the most cost-competitive, with NPC at 22,780,158\$, LCOE at 0.0668 \$/kW and annualized cost at 1.16 M\$.

Table 2. Cost Comparison between base case and proposed cases

Cases	CAPEX (M\$)	OPEX (M\$)	Annualized Cost (M\$)	NPC (M\$)	LCOE (\$/kW)
Case 0 (DG - PV - Grid)	3.23	1.20	1.37	26.9	0.0835
Case 1 (DG - PV - Batt)	2.97	1.62	1.77	34.8	0.109
Case 2 (DG - M PV - Batt)	3.77	1.41	1.60	31.4	0.0984
Case 3 (DG - M PV - Grid)	4.15	0.95	1.16	22.8	0.0668
Case 4 (DG-M PV-GRID-BAT)	4.18	0.95	1.17	22.9	0.0671
Case 5 (DG-PV-H2)	2.72	1.80	1.94	38.0	0.119
Case 6 (DG - M PV - H2)	3.97	1.53	1.74	34.1	0.107

C. Fuel Consumption and Emissions Analysis

This study highlights the necessity to reduce fuel use and emissions to address climate change. Table 3 shows that cases 4 and 3 are the most fuel-efficient in terms of annual carbon emissions. Notably, when the usage of DGs was reduced, there was a significant drop in emissions and fuel usage.

Table 3. Fuel Consumption and emissions Comparison between cases

Cases	Carbon Dioxide (Ton/yr)	Carbon Monoxide (kg/yr)	Total fuel consumed (Lit/yr)	Fuel (M\$)
Case 0 (DG - PV - Grid)	7,473.3	102	104,800	1.02
Case 1 (DG - PV - Battery)	7,927.7	2,908	2,998,000	29.4
Case 2 (DG - M PV - Batt)	6,738.9	2,472	2,548,000	25
Case 3 (DG - M PV - Grid)	6,379.3	82	84,267	0.83
Case 4 (DG-PV-GRID-BAT)	6,374.8	80	82,455	0.81
Case 5 (DG-PV-H2)	8,583.5	3,155	3,246,000	31.8
Case 6 (DG-M PV-H2)	7,119.5	2,623	2,692,000	26.4

VIII. CONCLUSION

HOMER PRO was utilized to mimic the current power system of the resort and evaluate six different power system scenarios to identify the most efficient power supply option. The primary criteria for selection are the attainment of a low LCOE and NPC, a minimal amount of excess electricity, and low carbon emissions. As a result of this study, analysis of all proposed cases revealed that hybrid on-grid system in case study 3, which adds 1.5 MW of PV capacity to the base scenario to reach 4 MW and provide 17.59 GWh/yr to meet the load, had the lowest LCOE of 0.0668 \$/kW, lowest NPC of 22.8 M\$, and lowest emissions. At 4.15 M\$, its CAPEX was the second highest, nevertheless. In contrast, case study six that included DG-PV-H₂ was a promising approach keeping in mind the rising costs of fuel and energy, coupled with inflation rates reaching record-breaking levels and the anticipated drop in the cost of H₂ system components in the future.

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